Morphology-Conditioned Physics-Based Motion Imitation Across 128 Body Shapes

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Abstract

Animating digital humans with different body proportions requires references consistent with each body's geometry and a controller that tracks them in physics. We build a paired corpus that links 20,951 motion identities to body-conditioned references and matching simulation assets for 128 body configurations, and train one morphology-conditioned temporal controller on 18,855 of these identities across all bodies. It achieves 99.14% tracking success on 1,048 held-out motions, with one selected trained body per clip. Replayed on 233 training-split clips across each of two panels of 128 unseen bodies, success is 97.28% inside the training coefficient range and 97.15% outside it, yet individual bodies succeed on as few as 3.0% and 13.3% of clips, and a later checkpoint improves outside-panel success while regressing individual bodies. In three-seed comparisons, attention variants reach higher mean success and lower position error than a temporal multilayer perceptron with less seed variability, while jerk differences stay within seed variation. An offline audit of 19,200 references shows that refinement reduces stance-foot sliding and below-floor body centres at the cost of higher reference jerk. The evidence supports strong average tracking and transfer within the evaluated domain; the causal contribution of explicit morphology inputs and the mechanism of the severe failures remain open.

Keywords: Physics-based animation; Digital humans; Motion imitation; Morphology-conditioned control; Temporal attention.

1 Introduction

Physics-based character animation should reproduce actions on digital humans with different proportions and dynamics. A shared controller offers a way to reuse a motion repertoire across those bodies, provided that references and simulated assets remain consistent. Changing limb lengths, collision geometry, and mass distribution changes both the spatial realization of a motion and the actuation needed to track it. The learning problem therefore has two coupled axes: variation in motion and variation in body. Good tracking on new motions for a familiar body does not establish transfer to a new body, and a high average over bodies can conceal severe failures.

Large motion resources, shape-conditioned reference generation, and scalable imitation provide the building blocks [1, 2, 3, 5, 7]. Body-conditioned control is established [10]. PHC and ASE learn from motion collections but report results on a single body shape; PHC supports body variation, although its quantitative evaluations use the mean SMPL body [5, 6] (Section 2.4). Our question is how reliably one controller learns and transfers across a large corpus of explicitly paired motions and bodies.

We approach it with a paired corpus in which every clip/body identity is preserved from HUMOS generation through simulation and replay: 20,951 clip identities and 128 body configurations,

on which one shared temporal slot/type-attention controller trains directly, using streamed motion residency, on 18,855 clips. Evaluation separates motion exposure from body exposure. The contributions are the paired corpus and workflow, the shared controller configuration with streamed residency, exposure-separated evaluation of trained and unseen bodies, and three-seed network comparisons together with a reference-refinement audit. We inherit the simulator integration, proximal policy optimization (PPO), and motion-tracking infrastructure from ProtoMotions; we do not claim these or morphology conditioning itself as new. HumanML3D captions are metadata, not policy inputs: this is reference-conditioned imitation, not text-conditioned control.

The original checkpoint achieves 99.14% success on 1,048 clip-disjoint test clips, each paired with one selected trained body. On 233 training-split clips across 128 unseen inside-range bodies and 128 unseen outside-range bodies, success is 97.28% and 97.15%. Yet the worst bodies succeed on only 3.0% and 13.3% of those clips. These findings support the central claim: *one shared controller learns morphology-specific tracking with strong average transfer to unseen shapes, but uneven robustness across individual bodies.* The design evidence explains practical trade-offs; it does not yet isolate the benefit of explicit morphology inputs or establish uniform held-out-motion performance across all bodies.

The remainder of the paper is structured as follows. Section 2 reviews related work; Section 3 defines the problem and system; Section 4 presents the corpus; Section 5 describes the proposed methodology; Section 6 specifies evaluation; Section 7 presents the results; and Section 8 discusses limitations and concludes. The appendices provide reproducibility details, training provenance, and additional unseen-body results; the complete per-body tables are in the Supplementary Material.

2 Related Work

2.1 Physics-based motion imitation

DeepMimic demonstrated reference-guided reinforcement learning for simulated character skills [4]. Perpetual Humanoid Control developed robust humanoid imitation for real-time simulated avatars [5]. These approaches motivate learning a feedback controller rather than directly replaying a kinematic trajectory. PULSE learns a motion representation for humanoid control, while OmniH2O, ExBody2, and GMT study large-scale control on robot platforms [17, 18, 19, 20]. ASE learns reusable latent skills and downstream task policies rather than requiring exact clip tracking [6]. These objectives provide context, but their published scores do not measure our clip/body replay task. We implement our experiments in ProtoMotions with Isaac Gym and PPO optimization [7, 8, 9].

2.2 Morphology-conditioned control and retargeting

Body-shape variation has already been studied in physics-based control: Won and Lee learned a parametric controller that accommodates variation in character height, weight, and proportions [10]. ShapeVerse trains physics-based characters with varied SMPL shape parameters to imitate walking and jogging [11]. A separate line of work learns policies that transfer across changes in agent *topology* – limb count and skeletal structure – through shared modular sub-policies [21], which is a different axis from the continuous anthropometric variation studied here. Morphology conditioning is therefore an established idea, and our work should not be interpreted as introducing it. We instead examine one controller over a large set of paired motions and bodies, with shape identity maintained throughout data generation, simulation, and evaluation.

Kinematic retargeting and dynamic tracking address different requirements. SMPL provides a common parameterized body representation [13], and SMPLSim constructs simulation-ready assets from these models [14]. Physics-based retargeting of motion capture to realistic human body shapes has been studied with spacetime optimization [12]. A shared skeleton makes poses comparable, but does not make global positions or required actuation independent of shape. Matching a reference to its intended simulation asset is consequently part of the task definition, not an optional evaluation convenience.

2.3 Text-motion data, shape-conditioned generation, and refinement

AMASS consolidates motion capture in a common body-model representation [1]. HumanML3D associates processed motion segments with natural-language descriptions [2]. We inherit those source identities and descriptions rather than claiming new motion capture or shape-specific language annotation.

HUMOS generates identity-conditioned human motion using a cycle-consistent learning formulation [3]. We use its pretrained model to obtain shape-specific references, not as the physics controller and not as a text-conditioned policy. The resulting sequences still require conversion to the simulator's skeleton and coordinate conventions, and their feasibility must be assessed in that simulator. Our refinement is a deliberately limited kinematic post-process; it is not a new generative model or a full dynamic trajectory optimizer.

The controller uses standard Transformer operations [15]. Learned slot and type embeddings identify temporal positions and observation sources. An ablation also investigates actor-only morphology modulation with adaLN-Zero, adapting the zero-initialized conditioning design used in Diffusion Transformers [16]. That experiment tests a conditioning mechanism, not diffusion-based action generation.

2.4 Comparison with related systems

Table 1 compares the closest systems on the axis studied here: the number of body shapes in their published results. PHC, ASE, and ProtoMotions learn from motion collections (11,313 AMASS training sequences for PHC, 187 clips for ASE), but their reported results use a single body shape. PHC evaluates on the mean neutral SMPL body; its appendix describes optional training on randomly sampled AMASS shapes, and shape variation is shown only qualitatively [5]. ASE reports experiments on one sword-and-shield humanoid [6]. ProtoMotions supports configurable characters and robots, but its documentation describes no single policy trained across body-shape variants [7]. Won and Lee directly establish parametric body control [10]; their experiments differ from our large clip-by-body imitation panels. Our distinction is therefore the combination of many motions and many body shapes under one controller, with references and assets paired per body and systematic inside- and outside-range replay, not priority for morphology conditioning.

This is a comparison of published capabilities, not a performance ranking. "Not reported" concerns the cited source and the specified unseen-body test, not impossibility. PHC is the closest reference-tracking comparator, but a matched comparison would require our shape-specific references, assets, clip splits, horizons, and success aggregation. ASE optimizes latent skills and downstream objectives rather than exact replay of designated clips, so its task scores are not interchangeable with tracking success. ProtoMotions supplies infrastructure and configurable trackers rather than a single published policy whose score can be compared directly with ours. These differences prevent direct use of published scores as matched baselines; they do not establish that the systems could not be adapted to this task. No external system has been rerun under our protocol.

Table 1: Body shapes in the published results of the closest systems.

System	Body shapes in reported results	Unseen-body evaluation
PHC [5]	One: mean neutral SMPL body	No systematic panel reported
ASE [6]	One: sword-and-shield humanoid	Not reported
ProtoMotions 3 [7]	One per configured character or robot	Not reported
Won & Lee [10]	Many: one parametric controller over height, weight, and proportions	Novel and changing shapes demonstrated
This study	Many: 128 trained bodies, one controller	128 inside and 128 outside the training range; all reported

Our quantitative design comparison is therefore internal, and no performance advantage over these systems is claimed.

Inherited functions and study additions. ProtoMotions provides the simulation abstraction, PPO training, motion-library and imitation infrastructure. HUMOS supplies the pretrained generator; SMPL/SMPLSim supply the body parameterization and asset construction. Our study adds the paired corpus assembly and identity checks, reference processing and fidelity audit, the evaluated temporal slot/type controller configuration, streamed clip residency, and exposure-separated evaluation. These are system and empirical contributions built on existing methods, not claims to invent attention, PPO, body models, or motion imitation. A directly comparable performance claim requires a matched external replay; none is made here.

3 Problem Formulation and System Overview

A body configuration $b = (g, \boldsymbol{\beta})$ specifies a SMPL template and shape coefficients. For source clip c, HUMOS supplies a shape-specific reference $\hat{s}_{c,b}(t)$ and SMPLSim supplies the matching asset. Changing b changes geometry, reference positions, mass/inertia, and mass-scaled actuation even though topology is shared. The task is to track these paired references with one policy, not impose identical world-space paths on all bodies.

We learn $\pi_\theta(a_t \mid o_t, b)$ using present state, history, future reference targets, and body information. The same parameters serve every body. Clips can change on reset while each environment retains its asset, and evaluation preserves reference/body identity. The flat-ground task has no external objects or partners, and captions are metadata, not controller input.

Figure 1 summarizes the system. Matching body parameters link generated references and simulation assets (Section 4); one shared controller trains on the resulting pairs with streamed motion residency (Section 5); and fixed pairs with known motion and body exposure are replayed for evaluation (Section 6).

4 Paired Motion and Body Corpus

We construct a derived corpus in which a source clip identity connects a motion description, 128 generated motion variants, and their matching simulation assets. A *clip* denotes a source segment, including a mirrored segment when present; a *variant* denotes that clip on one body configuration. Figure 1 shows the construction paths.

Motion	AMASS $\longrightarrow$ HumanML3D preprocessing and clip extraction $\longrightarrow$ HUMOS conditioned on $(g, \boldsymbol{\beta})$ $\longrightarrow$ MotionLib conversion and reference refinement $\longrightarrow$ per-clip motion variants.
Body	SMPL with the same $(g, \boldsymbol{\beta})$ $\longrightarrow$ SMPLSim $\longrightarrow$ matching skeleton, collision geometry, mass/inertia, and MJCF asset identity.
Text	HumanML3D source descriptions $\longrightarrow$ clip-ID lookup associated with the generated variants (metadata only; not a policy input).
The body assets are used for motion conversion, shape-specific forward kinematics, ground-clearance correction, and simulation. Refined reference motion and matching simulated state feed one shared policy.	

Figure 1: Paired reference and simulation-asset construction.

4.1 Source motion and body configurations

Motion. We use the AMASS-derived motion portion of HumanML3D [1, 2], drawn from the 18 source collections listed in Appendix A.1. Source SMPL+H sequences are processed at 20 Hz, with clip extraction, temporal cropping, mirroring, and treadmill/skating cleanup following the HumanML3D/HUMOS pipeline, so HumanML3D identities are preserved. The pipeline produced 22,459 candidate clip identities, counting original and mirrored clips; a text-based filter removed 1,508 that appear to require external objects or environmental support, leaving 20,951 identities for flat-ground imitation. The filter is a task-scope heuristic and does not prove that every retained reference is support-free or dynamically feasible. HumanML3D descriptions stay associated with the variants of their source clip as metadata; their semantic fidelity is not audited (Appendix A.2), and they supply keyword-based motion classes only for the exploratory analysis in Section 7.4.

Bodies. A body configuration is $b = (g, \boldsymbol{\beta})$, where g identifies the female or male SMPL variant and $\boldsymbol{\beta} \in \mathbb{R}^{10}$ contains shape coefficients. We sample

$$\beta_{ij} \sim \mathcal{U}(-3, 3), \qquad i = 1, \ldots, 64, \quad j = 1, \ldots, 10, \tag{1}$$

and cross the 64 vectors with the two variants, giving 128 body configurations rather than 128 independent coefficient vectors. Uniform sampling is a controlled source of variation, not a representative sample of human populations.

Reference and asset generation. For each clip, the pretrained HUMOS model [3] receives the processed pose and root features and the target body parameters, and returns a shape-conditioned reference (root-relative 6D rotations with translation). In parallel, SMPL bodies [13] with identical parameters are converted to MJCF assets with SMPLSim [14], using mass estimation and hand and foot simplifications. Each asset has 24 rigid bodies and 69 actuated joint coordinates, with its own collision proxies, joint ranges, and estimated masses and inertias. These are simulation-model estimates, not measurements of human participants.

4.2 Reference conversion and refinement

Generated sequences are converted to ProtoMotions' Z-up convention and resampled from 20 to 30 Hz. Forward kinematics (FK) uses the matching asset rather than a single neutral skeleton, and the packaged representation stores global body positions and rotations, local rotations, joint positions, velocities, and contact labels. Refinement is applied independently to each variant in five steps; default settings are in Table 10.

Table 2: Corpus and split sizes.

Partition	Clip identities	Motion variants
Training	18,855	2,413,440
Validation	1,048	134,144
Test	1,048	134,144
Total	20,951	2,681,728

Contacts. Ankle and toe contacts are validated, or detected from foot position and velocity when labels are unusable; short gaps are closed and very short runs removed. These label candidate stance intervals and are not measured forces. *Rotation filtering.* Joint rotations are filtered with sign-continuous quaternions, normalization, and a bounded geodesic correction; contact transitions divide the sequence into segments so that filtering does not average across a change of stance. *Root correction.* A conservative least-squares root translation, anchored at each contact's median foot position, reduces horizontal stance sliding. It is temporally regularized and capped, so it does not guarantee exact foot locking and does not solve for new joint angles. *Ground clearance.* With floor height zero and margin ϵ_z, the lift required at frame t is

$$\ell_t^{\text{req}} = \max\left(0, \epsilon_z - \min_{x \in \mathcal{C}_b(t)} x_z\right), \tag{2}$$

where $\mathcal{C}_b(t)$ is the collision geometry of body b, and a smoothed lift envelope stays above this requirement. This prevents penetration of the checked geometry in the reference; it does not establish balanced support, realistic impacts, or freedom from self-collision. *Reconstruction.* FK and velocities are recomputed from the corrected root and rotations within each motion.

The objective is modest: reduce measurable kinematic artifacts while preserving the generated motion. Section 7.3 reports artifact and fidelity measures; improved kinematics alone does not demonstrate improved policy learning or dynamic feasibility.

4.3 Scale and splits

Each per-clip file stores the 128 variants with frame counts, sampling rates, and the clip and asset identities needed to recover body parameters, so the training system can stream clip files without downloading the complete corpus. Splits are by clip identity: all variants of a clip, and original/mirrored partners, share a partition (Table 2), and variants are generated combinations rather than independent captures. The split is versioned with seed 42 and roughly 90/5/5 proportions; the 150 development clips used for design comparisons and the reference audit are reserved within the training split (Appendix A.7), and the test manifest is never loaded by the training workflow. The splits test held-out clip identities on the trained body set. They have not been shown to be disjoint by subject or capture recording, which limits claims about broader motion generalization.

5 Proposed Methodology: Shared Controller and Training

5.1 Task, simulation, and actuation

The policy follows the task in Section 3.

Experiments use the `smpl_mor` assets in Isaac Gym through ProtoMotions [8, 7]. Simulation runs at 60 Hz with two substeps and control decimation two, giving a 30 Hz policy rate. The policy

Table 3: Observation inputs supplied to both actor and critic.

Input	Dimension	Slots	Representation and path
Current state	$15B - 2 = 358$	1	Height; $3(B-1)$ positions; $6B$ rotations; $3B$ linear and $3B$ angular velocities. Current-state encoder.
History	358	7	Same state representation at the seven past offsets. Shared history encoder.
Future reference	$24B = 576$	4	Target and relative poses ($18B$), plus velocity differences ($6B$). Shared future encoder.
Historical action	$D = 69$	1	A raw previous policy-action vector, before tanh and PD scaling. Bypasses attention.
Morphology	11	1	$[g, \beta_1, \ldots, \beta_{10}]$ from the assigned asset. Bypasses attention.

outputs a raw action $a_t \in \mathbb{R}^D$, where $D = 69$. Joint targets are

$$q_t^{\text{target}} = q^{\text{offset}} + q^{\text{scale}} \odot \tanh(a_t), \tag{3}$$

with offsets and scales derived from the configured joint ranges. These targets drive the simulator's built-in position proportional–derivative (PD) controller. The action is not a residual added to the reference pose.

PD stiffness, damping, and effort limits are scaled per joint by the mass of its associated rigid body relative to that body in the reference asset. This is a body-level mass scaling, not a single scale based on total character mass, and it does not guarantee dynamically equivalent actuation across geometrically different bodies. Base gains and limits are listed in Table 12.

5.2 Observation design

The observation combines present state, history, future reference targets, a historical action, and explicit morphology information. Let $B = 24$ denote the number of simulated bodies. The state representation contains root height above ground, root-relative body positions excluding the root, 6D body rotations, and linear/angular velocities. Positions and velocities use a heading-aligned frame, and rotations remove the root heading. This removes dependence on arbitrary global heading and horizontal origin while retaining height and body configuration.

We include seven historical states at control-step offsets $\{1, 2, 3, 4, 8, 16, 32\}$, covering approximately 1.07 s, and four future reference slots at $\{1, 2, 4, 8\}$, covering approximately 0.27 s. Each historical state uses its own root/heading frame. Future slots contain target positions relative to the current root, heading-aligned target rotations, position differences from the current body state, relative rotations $R_{t,j}^{-1}\hat{R}_{t+k,j}$, and heading-aligned linear/angular velocity differences. Thus relative rotations are expressed with respect to the current body rotation, not as another heading-frame position feature.

The temporal context exposes both recent changes and more distant motion information, and the future targets provide a short anticipation window. Contact flags, an explicit phase scalar, text, and estimated physical feature vectors are not part of the selected observation.

Table 3 summarizes the actor and critic inputs. Dimensions are per slot. Positions use metres, linear velocities m/s, angular velocities rad/s, and 6D rotations are dimensionless. Raw actions and morphology coefficients are not joint-angle measurements.

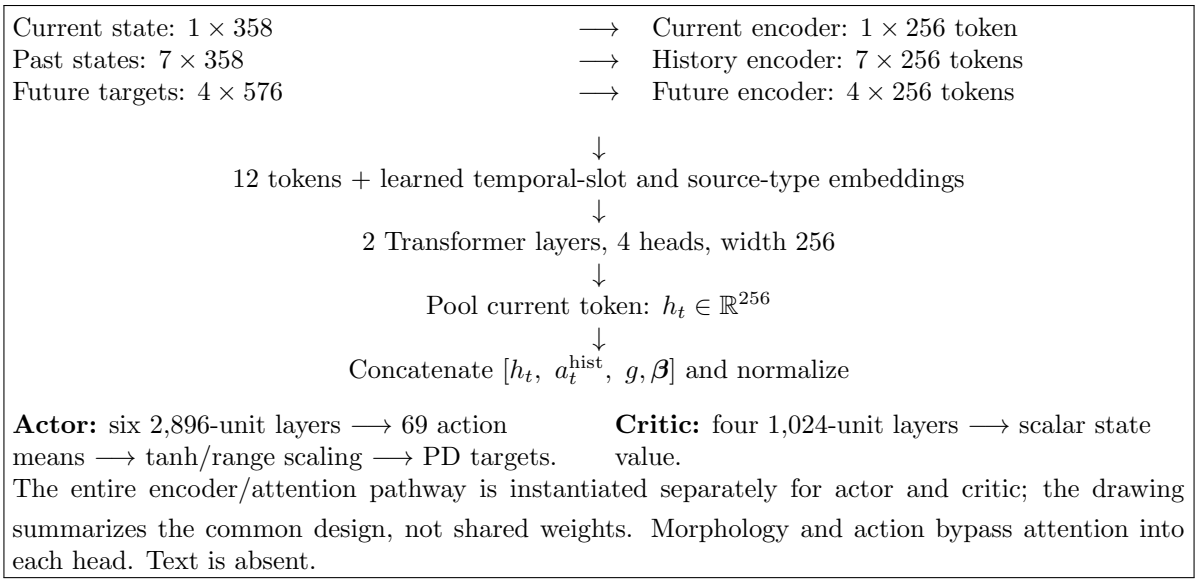


Figure 2: Temporal slot/type controller and separate actor/critic heads.

5.3 Temporal slot/type attention

We use three observation encoders: one for the current state, one shared across history slots, and one shared across future slots. Each is an MLP with two 256-unit ReLU hidden layers and a 256-dimensional token output. Inputs use running mean/standard-deviation normalization and a magnitude clamp of five.

The resulting 12 tokens comprise one current, seven historical, and four future tokens. For slot i, the Transformer input is

$$z_i = E_{\tau(i)}(x_i) + e_i^{\text{slot}} + e_{\tau(i)}^{\text{type}}, \qquad i = 0, \ldots, 11, \tag{4}$$

where $\tau(i)$ identifies the current/history/future source. Slot embeddings distinguish the configured temporal offsets, while type embeddings distinguish the observation sources. The Transformer has two layers, four attention heads, token width 256, and feed-forward width 1,024 [15]. Its current-state token supplies the pooled output after the configured output projection and activation.

Figure 2 traces the encoded temporal observations through attention and the separate actor/critic heads. The networks receive the same input categories but have separate parameters.

The final head input is the concatenation of the 256-dimensional attention output, the historical raw action, and the 11-dimensional morphology vector. This 336-dimensional input also uses running normalization and clamp five. Betas are not manually divided by three, embedded as a separate attention token, or used for adaLN modulation in the selected model.

Actor and critic have separate token encoders, attention modules, slot/type embeddings, and MLP heads. The actor head has six hidden layers of width 2,896; the critic has four of width 1,024, all with ReLU activations. The outputs are a Gaussian action mean and a scalar value, respectively. PPO samples actions with fixed diagonal log standard deviation -2.9; evaluation uses the mean action.

5.4 Tracking reward, resets, and PPO

The training reward combines global body tracking and root-height tracking. For corresponding simulated and reference body positions $p_j, \hat{p}_j$, rotations $R_j, \hat{R}_j$, and velocities $v_j, \hat{v}_j$ and $\omega_j, \hat{\omega}_j$,

define

$$e_p = \frac{1}{3B}\sum_{j=1}^{B} \|p_j - \hat{p}_j\|_2^2, \tag{5}$$

$$e_R = \frac{1}{B}\sum_{j=1}^{B} d_{\mathrm{SO}(3)}(R_j, \hat{R}_j)^2, \tag{6}$$

$$e_v = \frac{1}{3B}\sum_{j=1}^{B} \|v_j - \hat{v}_j\|_2^2, \tag{7}$$

$$e_\omega = \frac{1}{3B}\sum_{j=1}^{B} \|\omega_j - \hat{\omega}_j\|_2^2, \tag{8}$$

$$e_h = (h - \hat{h})^2. \tag{9}$$

where $d_{\mathrm{SO}(3)}$ is the geodesic rotation angle in radians. The coordinate-wise means in the position and velocity errors match the implemented reductions. The reward is

$$\begin{aligned} r_t = {} & 0.5\exp(-25e_p) + 0.3\exp(-5e_R) + 0.1\exp(-0.5e_v) \\ & + 0.2\exp(-0.1e_\omega) + 0.2\exp(-100e_h). \end{aligned} \tag{10}$$

The weights are not normalized to sum to one. There is no explicit effort, action-smoothness, or contact-matching reward in this experiment, and there is no source-dependent switch between canonical and HUMOS reward functions.

References are sampled only from variants matching the environment's asset. Reference-state initialization starts at frame zero with probability 0.2; otherwise it uses a random valid reference time. Training episodes have a maximum of 1,000 control steps and use the configured fall termination with a 0.15 m height setting and permitted foot-contact bodies. Tracking-error termination is disabled. This training rule is distinct from the evaluation failure threshold in Section 6.2.

We optimize using PPO [9], with 32-step rollouts, discount 0.99, GAE parameter 0.95, policy-ratio clipping 0.2, and normalized advantages. Separate Adam optimizers use learning rates 2×10^{-5} for the actor and 10^{-4} for the critic. Other configuration values and actual run budgets appear in Table 11 (Appendix A).

5.5 Direct multi-shape learning and streamed sampling

The final approach is direct multi-shape training on refined HUMOS references, with no neutral pretraining stage. The full-scale experiment streams per-clip files through `GlobalClipPool`: the training manifest is deterministically partitioned across GPU ranks, and each rank keeps a bounded resident pool containing all 128 variants of its selected clips, with environment-level motion sampling respecting the assigned morphology.

The pool is rebuilt every 256 epochs. Most resident slots are selected by priority, combining estimated clip difficulty and an exploration bonus, and a fraction 0.2 is reserved for uniform random selection to maintain rehearsal outside the highest-priority set. This is a training curriculum, not uniform sampling of the complete corpus. Validation membership is fixed independently of pool rebuilding and does not update sampling weights, so resident-pool and validation metrics measure different populations. Sampling constants and the pool size actually used are in Appendices A and B.

Table 4: Evaluation populations; each unseen panel has 128 bodies.

Evaluation	Motion exposure	Body exposure	Pairs
Validation / test	Held-out clips	Trained; one per clip	1,048 each
Development replay	150 training clips	All 128 trained	19,200
Unseen inside / outside	233 training clips	All 128 unseen per panel	29,824 each
Unseen diagnostic subset	23 val/test clips; seen by full-data	Same unseen panels	2,944 each
Trained-body control	150 training clips	All 128 trained	19,200
Design comparison	Development clips	Trained bodies	Three refined seeds

6 Evaluation Protocol

6.1 Populations and body-matched replay

Each reference is rolled out only in an environment using its exact asset identity, so a reference is never assigned to a different body. Evaluation uses the mean action and reference initialization, disables training termination, and records errors over the valid reference horizon, capped at 600 control steps (20 s at 30 Hz); success therefore refers to the evaluated horizon, not necessarily the whole source clip. Routine monitoring and the held-out evaluation select one body variant per clip with a deterministic hash of the clip identity (Appendix A), so the clip list is fixed while the body panel rotates across scheduled evaluations. Validation has been monitored repeatedly and is not an untouched test set; the clip-disjoint test manifest stays outside training and model selection and is evaluated once (Section 7.1).

The development corpus contains 150 clips and all 128 body configurations (19,200 variants), reserved within the full training split; it is not a held-out generalization benchmark. Because compute was limited, architecture and reference-processing choices are investigated on this subset with a fixed simulation budget, whereas the full-scale run trains on the 18,855 training clips (Table 2). Table 4 lists the populations used. Held-out identity is clip-grouped, not a demonstrated subject or recording split.

Unseen-body construction and replay. Each unseen panel contains 64 new coefficient vectors instantiated on male and female SMPL templates. Inside vectors lie in $[-3, 3]^{10}$; outside vectors have $\|\beta\|_\infty \in (3, 5]$, with 16 vectors (32 bodies) in each of four half-unit tiers. Inside denotes membership of the training coefficient box, not of the convex hull of trained bodies. Both panels use matching HUMOS references and assets, frame-zero offsets, and full reference refinement; the build report records zero penetration under its refinement diagnostic but does not certify dynamic feasibility. Replay uses mean actions, reference initialization, 32 environments per asset, and at most 210 control steps (7 s), with success as in Eq. (12). The 233 training-split identities isolate the body axis, while the other 23 validation/test identities remain separate. The trained-body control uses a different 150-clip panel and refinement run, so only within-dataset checkpoint deltas are paired comparisons.

Checkpoints and uncertainty. The original `refined` checkpoint is epoch 28,170 (distinct from the epoch-28,159 monitoring snapshot of Appendix B); the prespecified `fulldata` candidate is epoch 34,000 and trained on all original motion splits. Both are frozen for replay, and duration and motion coverage change together, preventing causal attribution. Each dataset has two original-checkpoint runs and one candidate run. Paired 95% bootstrap intervals resample clips with the

body panel fixed; they do not resample training seeds or unseen bodies. A change is called beyond noise only if its interval excludes zero and its magnitude exceeds the original-checkpoint rerun delta, so without a checkpoint ladder this measures simulator variability only. Per-body intervals are exploratory and unadjusted for multiple testing. On all 256 clips, identical-checkpoint success labels agree on 99.33% of inside and 98.68% of outside pairs, versus 99.96% in the control. Pooled success differs by at most 0.05 pp, but individual pairs can change by metres in peak error, so small pooled changes coexist with local instability.

6.2 Metrics

For each evaluated clip/shape pair i, the per-frame mean body position error is

$$d_{i,t} = \frac{1}{B} \sum_{j=1}^{B} \|p_{i,t,j} - \hat{p}_{i,t,j}\|_2, \tag{11}$$

and the configured success indicator is

$$S_i = \mathbf{1}\left[\max_{t<T_i} d_{i,t} \leq 0.5 \text{ m}\right], \tag{12}$$

where T_i is the evaluated horizon. This threshold is deliberately distinct from the smoother training reward, and passing it does not imply fine-grained pose accuracy or biomechanical realism. We also report time-averaged mean body-position error, mean global body-rotation geodesic error (not a local joint-angle error), the time average of the largest body-position error within each frame (the logged *mean maximum-joint error*), and the windowed normalized jerk of Appendix A.5, whose high-jerk statistic is the percentage of windows in which any body exceeds 6,500. Errors are averaged over valid frames per motion and then over motions, including failed rollouts; offline comparisons aggregate saved per-motion records over unique clip/shape pairs.

7 Results

We report held-out tracking, transfer to unseen bodies, and the design comparisons, and use per-body failures to delimit the claims.

7.1 Tracking across motions and bodies

Held-out motions on selected trained bodies. The epoch-28,170 checkpoint was re-evaluated offline with a standalone evaluator, once on the validation split and once on the clip-disjoint test split, under the protocol of Section 6: one selected body per clip (shape panel index 110), the exact asset for each reference, and mean actions. All 1,048 clips of each split last about 6.6 s (198 evaluated frames), so none reaches the 600-step cap. The test manifest is loaded only by this evaluator, which first verifies its hash, clip count, and disjointness from train and validation; the test split was evaluated once and not used to select the checkpoint.

Validation and test scores are similar (1,032 and 1,039 of 1,048 successes), and no significance claim is made from their difference. Continuous pose errors and jerk complement the permissive 0.5 m threshold. The Wilson intervals [22] are nominal: they condition on the checkpoint and selected body panel and ignore dependence between related recordings or mirrored clips, training variability, and uncertainty across body selections.

Table 5: Offline evaluation of epoch 28,170 on validation and test splits.

Metric	Validation	Test
Success rate (%) ↑	98.47	99.14
Nominal 95% Wilson interval (%)	[97.53, 99.06]	[98.38, 99.55]
Failure rate (gt err > 0.5) (%) ↓	1.53	0.86
Mean body-position error (m) ↓	0.0699	0.0676
Mean global body-rotation error (rad) ↓	0.1738	0.1723
Mean maximum-joint position error (m) ↓	0.1419	0.1393
Normalized jerk ↓	1,154.3	1,050.9
High-jerk windows (%) ↓	27.46	25.33
Worst single clip, gt err max (m)	3.79	1.67

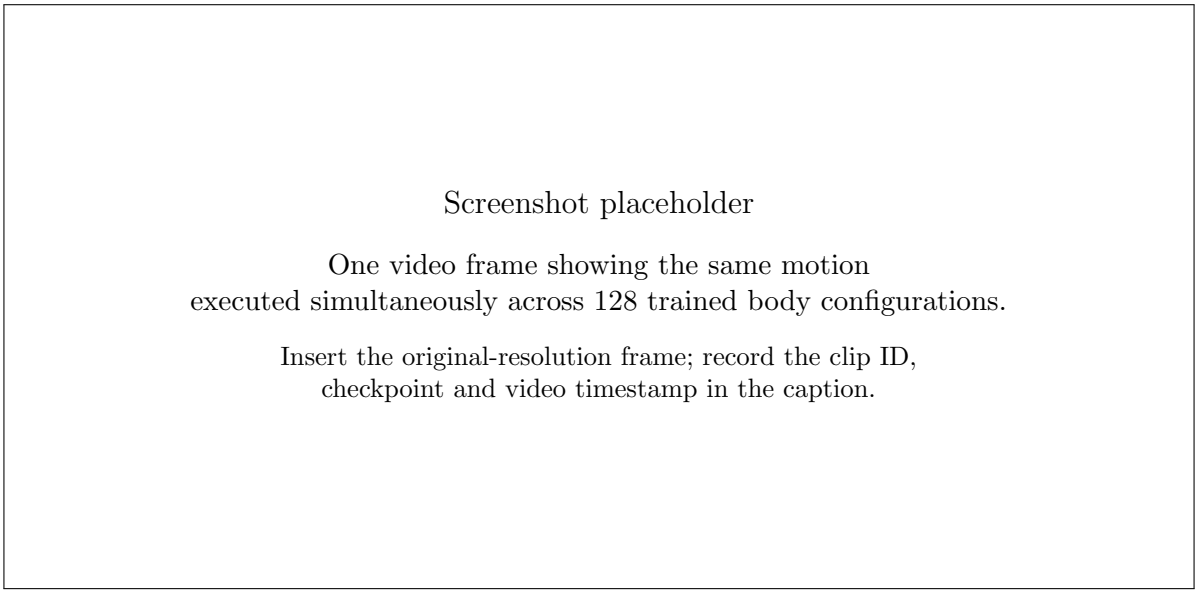


Figure 3: One motion executed across 128 trained body configurations (simulator collision geometry). Clip: [ID]; checkpoint: [identifier]; video time: [seconds].

Coverage of trained bodies. Replaying 150 training clips on all 128 trained bodies (19,200 pairs) reaches 99.29% success with the same checkpoint (Table 6), which establishes broad tracking on that subset. All-body consistency on held-out motions remains unmeasured: it would need a held-out-motion panel crossed with all 128 bodies. Figure 3 shows one motion executed across the 128 trained bodies; it illustrates execution and does not quantify tracking, contact quality, or transfer.

Residual failures. Nine test and 16 validation pairs fail the threshold. Their captions suggest ground transitions (six test, five validation failures) and whole-body rotations (two test, four validation), and also mention turning, side-stepping, an abrupt stop, and a hand-held sparring stick absent from simulation. These describe the failed sample rather than mechanical causes or class-specific failure rates; diagnosis would need synchronized rollouts and first-crossing traces, including successful variants of the same clips.

Table 6: Unseen-body success of the original and later checkpoints, with the trained-body control.

Panel	Refined (%)	Full-data (%)	Rerun Δ (pp)	Candidate Δ [95% CI] (pp)
Inside unseen	97.28	96.91	−0.02	−0.37 [−0.57, −0.18]
Outside unseen	97.15	97.57	+0.04	+0.42 [+0.22, +0.61]
Trained control	99.29	99.12	+0.02	−0.17 [−0.37, −0.03]

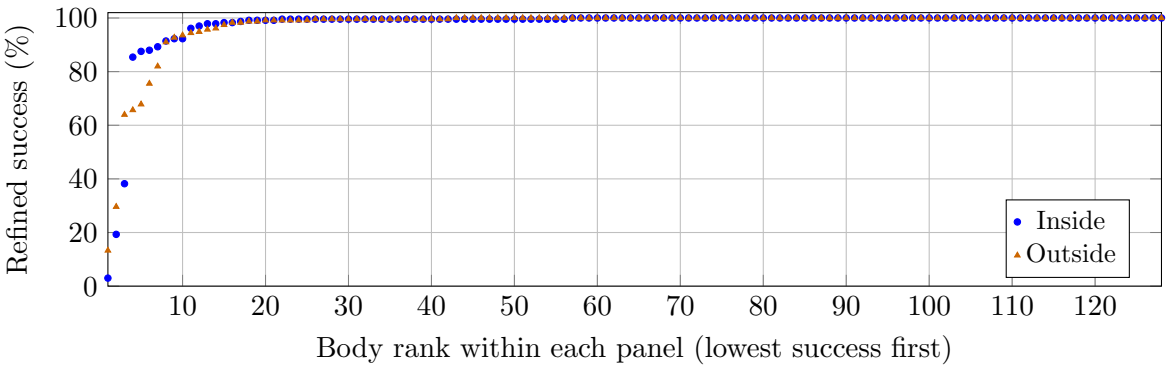


Figure 4: Unseen-body success, sorted within each panel.

7.2 Transfer to unseen bodies

High average transfer inside and outside the training range. Crossing 233 training-split clips with all 128 bodies in each unseen panel gives 29,824 pairs per panel, and Table 6 includes every body and every failed pair. The original checkpoint reaches 97.28% inside-range and 97.15% outside-range success. The trained-body control reaches 99.29%, but uses different clips and a different refinement run, so the gap of roughly two percentage points is descriptive, not a matched estimate of the effect of changing bodies. Deltas are paired within each dataset, intervals resample clips with the body panel fixed, and the rerun repeats the original checkpoint, not its training.

Failures concentrate in individual bodies. High averages coexist with severe body-specific failures. Inside body `male_669a0fb3` succeeds on 3.0% of clips and outside body `male_1c099619` on 13.3%; another inside body, `male_ec3bb7f8`, reaches only 19.3% despite an in-range mass of 42.7 kg and height of 1.41 m. Figure 4 shows all 128 bodies of each panel, sorted independently by original-checkpoint success on the same 233 known clips (the ranking does not pair inside and outside bodies), and the Supplementary Material reports every body. Lying outside the training coefficient box does not explain these failures: success does not decline monotonically across the four outside coefficient tiers (99.03, 97.76, 95.33, and 96.47%). Bodies outside the trained height range perform well on average (97.91% outside; 96.42% inside), while the tallest in-range bin is weaker (90.53% outside; 88.75% inside). The eight heaviest outside bodies, above 144.4 kg, average 79.02%, but small overlapping property groups and paired male/female templates are not independent samples of physical-property space, so coefficient extremity, mass, and height give no simple ordering of transfer difficulty (tier and mass tables in Appendix C). All failed rows have zero recorded termination flags; because flags depend on the replay configuration rather than an independent fall detector, we treat these as threshold failures pending synchronized rollouts and contact and root-height traces.

Table 7: Three-seed refined-reference comparison, mean ± s.d. across seeds.

Architecture	Success (%)	Pos. (m)	Rot. (rad)	Jerk	Params (M)
A: temporal MLP	95.14 ± 2.07	0.118	0.216	$2{,}110 \pm 643$	65.9
B: temporal attention	97.56 ± 0.34	0.090	0.218	$2{,}783 \pm 887$	51.2
C: slot/type attention	97.89 ± 0.22	0.090	0.219	$2{,}413 \pm 762$	51.2
D: C + adaLN-Zero	97.17 ± 0.08	0.083	0.211	$2{,}659 \pm 674$	52.1

Checkpoint sensitivity. The later full-data checkpoint (epoch 34,000) differs from the original by −0.37 pp inside and +0.42 pp outside (Table 6); both exceed the identical-checkpoint rerun change, but duration and motion coverage changed together, so the change cannot be attributed to either, and ordinary checkpoint variation remains unmeasured. The outside gain is concentrated in the ten bodies beyond the trained mass range (+7.68 pp, 95% CI [+6.39, +9.01]) and the outermost coefficient tier (+1.39 pp [+1.07, +1.72]), while mean position error rises by 2.4, 5.0, and 3.1 mm in the outside, inside, and control panels, so threshold success and continuous tracking move differently. Three inside and six outside bodies regress by more than 2 pp with intervals excluding zero, by up to 36.5 pp, and five outside bodies improve by more than 2 pp, including `male_1c099619` (13.3% to 71.2%) (Appendix C); neither improved extremes nor a higher outside average justify calling the later checkpoint uniformly better. On the 23-clip diagnostic subset, which the later checkpoint trained on, inside success is 97.08% versus 96.50% and outside 96.81% versus 96.91% for the original and later checkpoints; these values are excluded from the headline. Severe failures remain unresolved: per-asset reference feasibility and synchronized contact, root-height, and actuation traces have not been audited, so reference, controller, and asset mechanisms cannot be distinguished.

7.3 Design comparisons

We compare four architectures: A, temporal MLP; B, temporal attention; C, attention with slot/type embeddings; and D, C with actor-only adaLN-Zero. All retain morphology inputs in actor and critic. Refined-reference runs use three seeds (0, 1, 2) and a matched 6,000-epoch budget on the development subset, and each run contributes one summary, the mean of its last eight evaluations (epochs 4,600–6,000); means and sample standard deviations are computed across the three run summaries, and within-run evaluations are not replicates. Seed-1/2 values were checked against local TensorBoard records, whereas the original logs of the seed-0 values were not independently available. The full-scale model predates these reruns, which reassess its design rather than establish prospective selection. Cases E–H repeat A–D with unrefined references and one seed each (Appendix B).

Position and rotation errors are across-seed means, and parameters are actor and critic totals. Attention variants reach 97.2–97.9% mean success against 95.1% for the temporal MLP, with lower position error (0.083–0.090 versus 0.118 m) and much smaller seed variability (standard deviation 0.08–0.34 versus 2.07 points), although the MLP has the most parameters, roughly 26–29% more than the attention variants. Among the attention variants, C has the highest mean success and D the lowest position and rotation errors. A has the lowest mean jerk, but jerk differences between architectures are comparable to the seed-to-seed standard deviations (643–887), so the apparent factor-of-two jerk advantage in seed 0 is not a consistent architecture effect. Three seeds do not establish statistical superiority, and we retain C as a practical compromise.

Table 8: Reference-refinement audit over 19,200 development references.

	Unref. mean	Refined mean	Motions improved
Stance-foot speed (m/s) ↓	0.055	0.038	96%
Frames below floor (%) ↓	39.6	3.3	45%
Frames within 2 cm of floor (%) ↓	49.8	33.6	—
Position jerk (m/s^3)	45.2	60.7	21%
Per-body rotation change (deg)		0.12	
Root horizontal displacement (m)		0.008	
Cross-shape var., knee-flexion range (ratio r/u)		0.94	

Reference refinement. A paired offline audit of all 19,200 development references (150 clips × 128 bodies; no simulation) compares artifact and fidelity measures before and after refinement (Table 8). Artifact rows give the corpus mean before and after and the fraction of motions that improve; fidelity rows give the corpus mean refined-versus-unrefined difference. Refinement lowers stance-foot speed during detected contact from 0.055 to 0.038 m/s, with 96% of references improving, and below-floor frames from 39.6% to 3.3% (only 45% of references improve on the latter because many were already grounded). Measured motion features are largely preserved: the mean per-body rotation change is 0.12°, root horizontal displacement is 0.008 m, and the cross-shape variance of knee-flexion range is retained at a ratio of 0.94. The cost is higher reference jerk (the raw mean third difference of body positions, not the windowed evaluator score), from 45.2 to 60.7 m/s^3, with 79% of references worsening; temporal root and clearance corrections are a candidate explanation, but their contribution to policy smoothness is not isolated. Refined and unrefined policies (cases A–D versus E–H) track different targets, so their comparison describes different tasks and does not establish a causal benefit of refinement, which would need a common-reference checkpoint comparison.

Scope of the morphology-conditioning evidence. Every architecture receives morphology inputs in actor and critic. The adaLN-Zero variant changes how the actor uses them but does not test their removal, and body-dependent state and reference targets also expose morphology. Successful transfer therefore establishes a capability of the whole system, not that explicit coefficients cause it. Attributing it to those inputs needs a matched training run without them, with the same references, body panel, budget, and seeds; inference-time masking or shuffling would introduce distribution shift and cannot replace that comparison.

7.4 Exploratory physical analysis

As a secondary question, we ask whether cross-body variability in control demand differs between dynamic and quasi-static motions. The original checkpoint is replayed over the 150 development clips and all 128 trained bodies (19,200 pairs, zero recorded failed episodes under this analysis's own diagnostic). Keyword matching on the inherited captions assigns running, jumping, and kicking to a dynamic group (25 clips) and standing and sitting to a quasi-static group (37 clips); the other 88 clips are excluded, and the labels have not been manually validated. After subtracting each clip's across-body mean, Table 9 reports the standard deviation of the pooled residuals, with 95% intervals that resample clips as clusters (10,000 resamples). Torque, power, and jerk describe the simulated controller, not calibrated human biomechanics.

The spreads differ modestly and their separate intervals overlap, which is not a test of the

Table 9: Cross-body control-demand variability, dynamic versus quasi-static clips.

Metric	Dynamic std [95% CI]	Quasi-static std [95% CI]
Normalized torque (frac. effort limit)	1.93 [1.71, 2.21] $\times 10^{-4}$	2.06 [1.86, 2.30] $\times 10^{-4}$
Mass-normalized power	0.292 [0.220, 0.370]	0.259 [0.210, 0.323]
Normalized jerk	350.2 [283.3, 416.9]	403.2 [303.3, 513.5]

class-level difference. A direct dynamic-minus-quasi-static contrast with clip-level resampling or permutation has not been estimated, so the analysis supports descriptive comparisons only, not a positive or negative interaction result.

8 Discussion and Limitations

We studied shared physics-based character animation through paired morphology-specific references and assets, a temporal controller, and explicit separation of motion and body exposure. One policy trained across 128 bodies achieves 99.14% success on held-out motions with one selected trained body per clip, and executes development clips across all trained bodies. It transfers to unseen bodies at 97.28% inside and 97.15% outside the training coefficient range, but individual bodies can fail on most clips, and a later checkpoint introduces both recoveries and regressions. Temporal attention improves tracking accuracy and seed stability over a temporal MLP without a consistent smoothness difference, and the reference audit shows reduced contact artifacts at the cost of higher reference jerk. These comparisons support practical architecture and reference-processing choices, but not a causal explanation of transfer.

Implications for character animation. The evaluated workflow combines body-specific references with one shared feedback controller, supporting reuse across the tested digital-human configurations. The reference audit and the simulated-motion results are complementary measurements: reference cleaning does not establish lower sliding or penetration in controller outputs, which remain unassessed, and high threshold success is tracking capability, not a complete assessment of visual realism.

Average transfer and individual robustness. Similar inside and outside averages do not make body extrapolation uniformly reliable. Success is non-monotonic across coefficient tiers, and poor tracking occurs both beyond and within the training mass and height ranges; these associations describe the tested assets and do not identify failure causes. The later checkpoint recovers some bodies and harms others, so threshold success alone would overstate its improvement.

Coverage and inference. The unseen-body headline uses known motion identities to isolate body transfer, the 1,048-clip test uses one trained body per clip, and a fixed held-out-motion panel crossed with all 128 trained bodies remains missing. Original and mirrored variants stay together, but the splits are not shown to be subject- or recording-disjoint. Clip-bootstrap intervals hold the body panel fixed and do not quantify uncertainty over shape space, and one full-scale run with one candidate checkpoint does not measure training-seed or ordinary checkpoint variation.

Mechanism and design. State and reference observations already carry shape information, so attributing transfer to explicit morphology inputs needs a matched policy without them. The

architecture comparison keeps those inputs in every model and has unequal capacities, with the temporal MLP the largest. Refinement reduces contact artifacts with small pose changes but raises target jerk, and policies evaluated against different targets do not isolate its learning benefit. The exploratory physical analysis (Section 7.4) remains descriptive.

Failure diagnosis and scope. Failures are error-threshold failures with zero termination flags, which alone do not prove the absence of falls. Synchronized rollouts, contact and root trajectories, and per-asset feasibility audits are needed before assigning a mechanism; extreme HUMOS references have been refined but lack such validation. Estimated masses, collision proxies, and mass-scaled gains are modeling choices, and tracking success and jerk do not certify biomechanical realism. The short, reference-initialized flat-ground protocol excludes objects, partners, long-horizon recovery, skeletal-topology changes, and real robots. The corpus is derived, so clip/body variants are not independent captures, and caption alignment is not audited. A matched external tracker would strengthen comparative conclusions, but the published capability comparison alone cannot establish a performance advantage.

Summary and outlook. The evidence supports strong average shared tracking and transfer within this protocol. All-body held-out-motion coverage, matched conditioning and external controls, and dynamic diagnosis of severe failures remain necessary for stronger claims about robustness and mechanism.

Statements and Declarations

Funding [Funding sources and grant identifiers, or a statement that no funding was received.]

Competing interests [Authors to declare any financial or non-financial competing interests, or confirm that none exist.]

Author contributions Hansen Han: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, visualization, writing—original draft, writing—review and editing, supervision, project administration, and funding acquisition. This provisional allocation will be confirmed and amended by the authors before submission.

Data and code availability The motion corpus derives from AMASS and HumanML3D and remains subject to the source datasets' access and licensing conditions. The research code repository is https://github.com/hansen1416/ProtoMotions; demonstrations are available at https://hansen1416.github.io/morph_mimic/. Evaluation summaries and per-body results used in this manuscript are retained with the research records. [Authors to specify the archival code version, evaluated checkpoints and assets, evaluation-data access location, and any restrictions on derived data.]

AI assistance in manuscript preparation An AI assistant was used to help restructure and draft manuscript text and check consistency with the recorded sources. Reported experimental measurements come from existing research records; this assistance did not produce new training or rollout results.

A Reproducibility Details

A.1 Source preparation and refinement settings

The AMASS source collections used by the preparation pipeline are ACCAD, BioMotionLab_NTroje, BMLhandball, BMLmovi, CMU, DFaust_67, EKUT, Eyes_Japan_Dataset, HumanEva, KIT, MPI_HDM05, MPI_Limits, MPI_mosh, SFU, SSM_synced, TCD_handMocap, TotalCapture, and Transitions_mocap. Source preparation operates at 20 Hz; the final motion libraries use 30 Hz. Joint ordering and quaternion conventions must follow the matching asset and MotionLib conversion rather than be inferred from source array positions. Body coefficients are sampled with NumPy's random number generator using seed 46, and bodies are identified by stable beta keys rather than content hashes. SMPLSim assets use mass estimation and hand and foot simplifications (`real_weight`, `replace_feet`, `remove_toe`, `freeze_hand`).

Table 10: Default settings of the current reference-refinement procedure. Frame counts refer to the 30 Hz MotionLib representation. These are kinematic-processing settings, not physical constraints learned by the policy.

Operation	Setting
Contact candidates	Left/right ankle and toe bodies
Contact detection thresholds	Speed 0.15 m/s; height 0.10 m
Contact cleanup	Close gaps up to 2 frames; remove runs shorter than 3 frames
Rotation filter	Savitzky–Golay, 9-frame window, polynomial order 3
Rotation blend / bound	Strength 0.5; maximum geodesic correction 8°
Contact-transition guard	Separate segments; 2-frame transition taper
Root XY correction	Median stance anchors; least-squares common translation
Root correction regularization	21-frame smoothing window; 0.10 m displacement cap
Ground clearance	0.005 m collision-geometry margin; 5-frame lift smoothing
Final reconstruction	Per-asset FK and per-motion velocity recomputation

A.2 Caption provenance

HumanML3D descriptions are available through a separate clip-ID lookup and the annotation metadata used by HUMOS. A description is associated with the variants of its source clip; it is not a new annotation for each generated body. A coarse check against a single-caption-per-clip lookup finds a caption for all 20,951 clip identities, but semantic fidelity is not established: of the 9,579 mirrored identities, 22% carry a caption byte-identical to their un-mirrored source and 40% contain an explicit left/right or clockwise term that mirroring inverts, so removing a mirror prefix does not establish a correct caption. Segment timing and multi-annotation coverage against the full HumanML3D annotation set are not audited, and the keyword-based motion classes used in Section 7.4 have not been manually validated. The retained and excluded ID lists provide the audit trail for the object-support filter.

A.3 Training configuration

Table 11 records the shared optimization settings and the environment and batch sizes actually used; run-specific resolved configurations take precedence over older launch examples. The full-scale frame count is a recorded stopping snapshot, not a budget matched to the development runs. Clip priority combines estimated difficulty, updated from training-pool evaluations with an exponential

Table 11: Training configuration; environment and minibatch counts are per GPU.

Setting	Corrected 150-clip runs	Full-scale selected run
Parallel GPUs per policy	1	6
Environments per GPU	4,096	2,048
PPO minibatch size per GPU	16,384	8,192
Rollout length	32 control steps	32 control steps
Mini-epochs per update	1	1
Actor / critic learning rates	2×10^{-5} / 10^{-4}	2×10^{-5} / 10^{-4}
Discount / GAE / ratio clip	0.99 / 0.95 / 0.2	0.99 / 0.95 / 0.2
Gradient norm clip	50	50
Evaluation cadence	200 epochs	256 epochs
Training seed	0–2 (refined A–D); 0 (unrefined E–H)	0 (saved run configuration)
Reference residency	All 150 clips	256, later 128 clips per GPU
Training frames	786,432,000 per run (6,000 epochs)	Approximately 11.07×10^9 recorded
Status of reported evidence	Completed, Table 7	Epoch-28,170 checkpoint, Table 5

moving average of rate 0.1 and a floor of 0.05, and an exploration coefficient of 1.0; the pool is rebuilt every 256 epochs and a fraction 0.2 of slots is reserved for uniform selection.

A.4 PD gains and limits

Table 12 gives the base values configured for the SMPL morphology robot. For joint d belonging to rigid body j, the simulator applies the multiplier

$$\rho_{b,j} = \frac{m_{b,j}}{m_{\mathrm{ref},j}} \tag{13}$$

to stiffness, damping, and effort limit. The velocity limit is not mass-scaled. All relevant asset masses, joint ranges, collision settings, and the reference asset must accompany a reproducible physical interpretation.

Table 12: Base PD and actuation values before per-body mass scaling. Stiffness and damping use the simulator's rotational PD units (Nm/rad and Nm s/rad); effort limits are in Nm and velocity limits in rad/s.

Joint family	Stiffness	Damping	Effort limit	Velocity limit
Hip, knee, ankle	800	80	500	100
Toe	500	50	500	100
Torso, spine, chest	1,000	100	500	100
Neck, head, thorax, shoulder, elbow	500	50	500	100
Wrist, hand	300	30	500	100

A.5 Windowed jerk convention

The logged smoothness score is computed from simulated global body positions using consecutive finite differences with control interval $\Delta t = 1/30\,\mathrm{s}$. A nominal 0.4 s window contains $W = \max(4, \lfloor 0.4/\Delta t \rfloor)$ frames, giving $W = 12$ at the configured rate and duration $T_w = (W-1)\Delta t$. For

body j in window w, let $L_{w,j}$ be the sum of speed times Δt and let $\dddot{p}_{w,j}$ denote the third finite difference. The implementation uses

$$J_{w,j} = \frac{T_w^5 \sum_{t \in w} \| \dddot{p}_{t,j} \|_2^2 \, \Delta t}{\max(L_{w,j}, \epsilon)^2 + \epsilon}, \qquad \epsilon = 0.1, \tag{14}$$

where the sum includes the valid third-difference samples in the window. Window scores are averaged per body and then across bodies and motions. High-jerk percentage counts windows for which $\max_j J_{w,j} > 6{,}500$.

Equation (14) documents the existing numerical convention, including its regularizers; it is not presented as a universally dimensionless or sampling-rate-invariant biomechanical measure. Comparisons must keep the sample rate, window, units, regularizers, and aggregation fixed. Reference refinement reports also include raw jerk in $\mathrm{m/s^3}$; those values are a different metric and cannot be compared numerically to this logged score.

A.6 Evaluation panel selection

Routine monitoring selects one variant per clip. With seed $s = 42$, stable clip identity c, panel index p, and N_c available variants, the choice is

$$k(c, p) = \left(\mathrm{uint64}_{\mathrm{big}}\big(\mathrm{SHA256}(s\!:\!c)_{0:8}\big) + p\right) \bmod N_c. \tag{15}$$

Here $s\!:\!c$ is the UTF-8 string `seed:clip_id`. For scheduled monitoring, p is the integer quotient of the checkpointed epoch and the evaluation cadence, which gives deterministic staggered shape choices that rotate across evaluations and remain reproducible after resume. The validation clip list is fixed while the selected shape panel changes, so successive scores are not repeated measurements on an identical set of clip/shape pairs. The standalone validation and test evaluations use panel index 110.

A.7 Split and experiment provenance

The split metadata is stored in `data/splits/hhi_stage2_v1/split_metadata.json`, with schema version 1, split seed 42, and split identifier

```
4c1c9d592e3dbeef15abe18cb50a3141820735849414ecc2d8959e9f38502263
```

It records the source-manifest hash, training/validation/test manifest and ID-list hashes, difficulty information, and the forced training reservation. The generator uses 100 difficulty strata. Its forced training reservation contains the 150 requested development identities and their existing mirrored partners, totalling 286 identities. This is a split-assignment count, distinct from the 150-clip development corpus used for the reported audit and design comparisons. Checkpoint/run split provenance refers to these artifacts, not just a verbal 90/5/5 ratio.

The documented HUMOS configuration selects `logs/humos/q6zbv2tu/checkpoints/latest-epoch=1599.ckpt`. The immutable HUMOS checkpoint checksum and the SMPLSim / asset-generation revision hashes are to be archived with the dataset release; a checkpoint path identifies the local workflow but is not a substitute for a checksum.

The principal implementation entry points are:

- Reference refinement: `tools/refine_humos_motion.py`; streamed per-clip processing: `tools/refine_humos_r2_per_clip.py`.

- Selected full-scale experiment: `examples/experiments/mimic/mlp_wide_stage2_discover_attention_slot_type.py`, inheriting the common attention experiment and enabling slot/type embeddings.
- Corrected development comparison: `tools/train_150_evalfix_ablation.sh` (cases A–E) and `tools/train_150_evalfix_unrefined.sh` (cases F–H), seed 0.
- Motion residency and evaluation: `protomotions/components/global_clip_pool.py` and `protomotions/agents/evaluators/mimic_evaluator.py`.
- Standalone split evaluation: `evaluate_validation_split.py` and `evaluate_test_split.py` under `protomotions/`. Each output records the checkpoint, split, reference version, and shape panel; the test evaluator additionally checks the test-manifest hash, count, and disjointness before running.

A.8 Unseen-body and repeated-seed evidence

The sole unseen-body study is the inside/outside replay of 9 October 2026. Its source report is `r2:proto-data/unseen_shapes_v2/eval/REPORT.md`; a local read-only snapshot is retained under `results/paper_evidence/unseen_shapes_v2/`, including both 128-row per-shape CSVs, comparison summaries, checkpoint identities, and rerun checks. The evaluator is `tools/analyze_unseen_morphology_generalization.py`; paired analysis uses `tools/compare_checkpoints_unseen_morphology.py`. The report records code revision `75324d9` at analysis time.

Frozen checkpoint SHA-256 values are:

```
refined (epoch 28,170):
4b83f20c93bf95fab389fc8935685ef8a2c7e46583f765e8c802421edb4479cb
fulldata (epoch 34,000):
7b57de8ec047c390c2881f8a630d7196a09da6dd08bfbae743f36d6542cd2192
```

Each checkpoint has its own frozen resolved inference configuration. All per-pair outputs and logs remain on R2; the local small-artifact snapshot is not a copy of the complete rollout records. The 233/23 clip partition, body manifests, and physical-property groups are defined in the source report.

Seed-1/2 architecture records are in the corresponding `results/hhi_150_evalfix_A_seed1` through `results/hhi_150_evalfix_D_seed2` run folders. Resumed logging segments must be joined by evaluation epoch before forming last-eight summaries. Across-seed standard deviation is computed from three run summaries, rather than from all scheduled evaluations pooled as independent observations.

For each final experiment, retain the resolved configuration, code revision, model/checkpoint checksum, body coefficient and asset manifests, split hashes, all resume segments, frame count, GPU model/count, and elapsed GPU-hours. For each evaluation, retain per-pair clip and asset IDs, panel selection, success/failure, valid horizon, and component errors in a local analysis file. The main full-scale W&B identifier is `3mvgad6f`; the corrected small-scale group is `hhi_150_evalfix_comparison`. The evaluation/sampling correction is recorded in commit `3685247`.

Immutable HUMOS-checkpoint and SMPLSim-revision checksums and public release of the retained validation/test per-motion records remain incomplete. The present study therefore does not claim a fully released benchmark.

Table 13: Recorded full-scale monitoring snapshot.

Metric	Hard resident pool	Validation holdout
Success rate (%) ↑	89.84	98.47
Mean body-position error (m) ↓	0.0960	0.0698
Mean global body-rotation error (rad) ↓	0.2261	0.1760
Mean maximum-joint position error (m) ↓	0.1848	0.1412
Windowed normalized jerk ↓	1,442.8	1,151.7
High-jerk windows (%) ↓	31.05	27.25

B Development and Training Provenance

The tracking results use the original epoch-28,170 checkpoint; the unseen-body results also compare the frozen epoch-34,000 candidate. The monitoring snapshot below is a separate measurement at epoch 28,159. Historical single-seed comparisons and the full-scale run record are retained here to distinguish provenance from final performance.

B.1 Recorded full-scale validation result

Table 13 reports the final scheduled monitoring evaluation recorded for the selected full-scale slot/type run, at epoch 28,159 after approximately 11.07 billion simulated frames. The separately downloaded checkpoint is from epoch 28,170. These values are post-correction, morphology-matched evaluations, not measurements of that later checkpoint or of the clip-disjoint test split.

Recorded full-scale monitoring snapshot. Validation contains 1,048 held-out clips; each evaluation uses one selected trained body per clip. The active resident pool is difficulty-biased. Values are the stored distributed logging summaries from run `3mvgad6f`, epoch 28,159; no confidence intervals or all-shape coverage are implied.

The validation score is higher than the resident-pool score, while the reported errors are lower. This comparison is consistent with a curriculum that concentrates residency on difficult clips: the resident pool is not an independent uniform sample of the training distribution. Its 89.84% success rate should not be interpreted as the success rate of the entire training corpus, nor should the difference be described as validation outperforming an IID training sample.

Figure 5: Scheduled success rate over training for the full-scale slot/type run (`3mvgad6f`). The step near epoch 16.9k is the morphology-matched evaluation correction plus its sampling change, not an instantaneous learning event; pre-correction points use the earlier, sometimes mismatched reference-to-asset assignment and are not comparable to the post-correction curve. After the correction the holdout success rate climbs from about 0.96 to a stable 0.98–0.985. The resident-pool curve is noisier because residency is continually re-concentrated on hard clips.

The 98.47% result supports strong tracking of the selected held-out clip/body pairs under the specified threshold. It does not establish all-body success for those clips, unseen-body generalization, or an untouched test-set score. The thresholded success rate is complemented by position, rotation, and smoothness measures because a successful rollout can still have visible local errors or jitter.

The run record is available at the full-scale W&B run. The current table uses the recorded snapshot, not a newly executed evaluation.

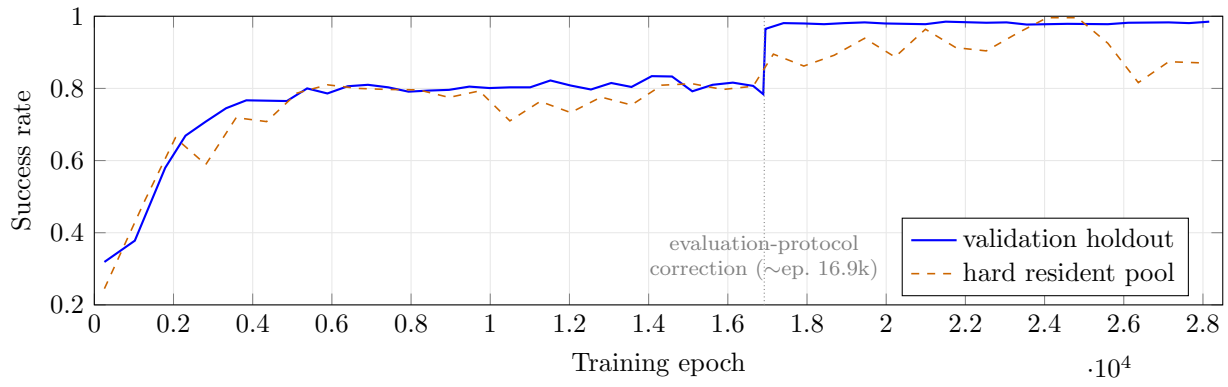


Figure 5: Scheduled success during full-scale training.

Table 14: Historical seed-0 development grid. Success uncertainty is variation across eight evaluations within each run. Other columns are their means. Refined and unrefined policies track different targets. The three-seed comparison is in Table 7.

Case	Architecture / data	Success (%)	gt err	gr err	Jerk	HJ%
A	temporal MLP, refined	97.5 ± 0.8	0.109	0.207	1,729	33.3
B	attention, refined	97.5 ± 1.4	0.098	0.228	3,455	40.4
C	slot/type, refined	**98.1 ± 0.6**	0.097	0.226	**1,757**	34.2
D	slot/type + AdaLN, refined	97.1 ± 1.2	**0.090**	0.220	3,145	38.9
E	slot/type, unrefined	97.9 ± 0.6	0.102	0.219	1,422	29.6
F	temporal MLP, unrefined	96.2 ± 1.6	0.115	0.212	2,927	36.6
G	attention, unrefined	97.8 ± 1.2	0.098	0.219	3,117	34.9
H	slot/type + AdaLN, unrefined	97.5 ± 0.9	0.091	0.207	2,956	34.0

B.2 Historical single-seed reference grid

Cases A–D use refined references and E–H use unrefined references. Within-run variation must not be interpreted as uncertainty across training runs. The main architecture conclusion uses repeated refined runs instead.

B.3 Full-scale run record

The full-scale selected run shares the architecture of case C exactly (same encoder, transformer, and head widths; `mlp_wide_stage2_discover_attention_slot_type.py` inherits these constants unchanged from the 150-clip experiment file), so its parameter count is also 51,238,582. Parameter counts for the four architectures (Table 7) are computed analytically from each run's resolved model configuration, because every linear layer is instantiated with `nn.LazyLinear` and has no fixed count outside a materialized model. The run used six A40 GPUs (`ddp`, 32-bit, 2,048 environments and an 8,192 minibatch per GPU) for roughly 28,170 epochs; the logged rank-0 wall time is about 77 h, on the order of 450 GPU-hours. An early restart after a memory-related failure resumed from the last checkpoint with the resident pool reduced from 256 to 128 clips per rank and the local cache multiplier set to 1.0; because configurations are reloaded from the pickle on resume, the frozen configuration still records the pre-restart pool size of 256. The earlier failed launch was logged into the same run.

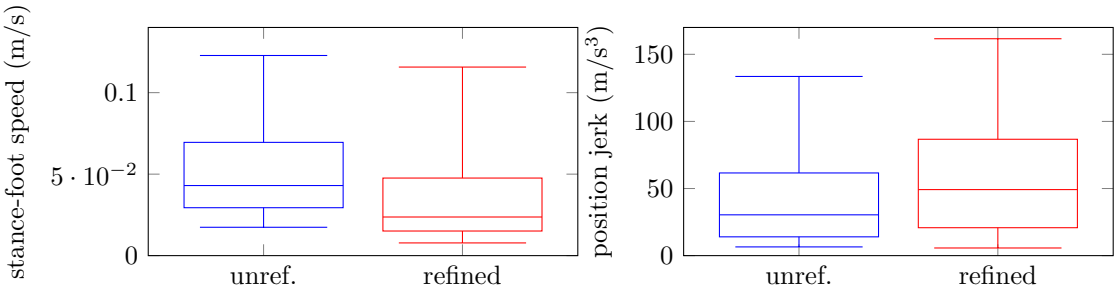


Figure 6: Reference sliding and jerk distributions before and after refinement.

Table 15: Outside coefficient tiers and simulated-mass bins, 233-clip panel.

Group	Bodies	Refined (%)	Full-data (%)
$\|\beta\|_\infty \in (3, 3.5]$	32	99.03	99.46
$\|\beta\|_\infty \in (3.5, 4]$	32	97.76	97.81
$\|\beta\|_\infty \in (4, 4.5]$	32	95.33	95.12
$\|\beta\|_\infty \in (4.5, 5]$	32	96.47	97.87
Mass $<$ 26.4 kg	2	100.00	100.00
Mass 26.4–144.4 kg	118	98.33	98.13
Mass $>$ 144.4 kg	8	79.02	88.62

B.4 Reference refinement distributions

Figure 6 shows per-motion distributions (box: quartiles; whiskers: 5th–95th percentile) over the 19,200 development references. Stance-foot speed drops across almost the whole distribution, while position jerk rises across almost the whole distribution.

C Additional Unseen-Body Results

This appendix adds the coefficient-tier and mass-bin summary and the checkpoint-regression list that support the unseen-body results. The complete per-body results for both panels, 256 rows in total, are in the Supplementary Material.

C.1 Coefficient tiers and mass bins

Table 15 reports the predefined outside coefficient tiers and descriptive simulated-mass bins on the 233-clip panel. Tier and mass rows overlap, each partition covers all 128 bodies, and the mass thresholds are the trained-asset range. These are associations, not identified causes.

C.2 Checkpoint regressions

Table 16 lists all bodies whose paired success-change interval lies entirely below -2 percentage points on the 233-clip panel. These exploratory per-body intervals are not adjusted for multiple comparisons.

Table 16: Bodies with a paired success decrease beyond 2 percentage points.

Panel	Asset	Refined (%)	Full-data (%)	Δ (pp)
Inside	female_e7bc8c92	92.3	79.0	−13.30
Inside	male_6d81a8f6	97.9	85.0	−12.88
Inside	male_ec3bb7f8	19.3	10.7	−8.58
Outside	female_edcf2449	94.8	58.4	−36.48
Outside	male_a7f6b877	65.7	40.8	−24.89
Outside	male_edcf2449	82.0	67.0	−15.02
Outside	female_e706da88	94.4	79.8	−14.59
Outside	female_bd0ec1ce	63.9	52.8	−11.16
Outside	female_f9bf3075	97.4	88.8	−8.58

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